Type of Vegetarian Diet, Body Weight, and Prevalence of Type 2 Diabetes

OBJECTIVE We assessed the prevalence of type 2 diabetes in people following different types of vegetarian diets compared with that in nonvegetarians. RESEARCH DESIGN AND METHODS The study population comprised 22,434 men and 38,469 women who participated in the Adventist Health Study-2 conducted in 2002–2006. We collected self-reported demographic, anthropometric, medical history, and lifestyle data from Seventh-Day Adventist church members across North America. The type of vegetarian diet was categorized based on a food-frequency questionnaire. We calculated odds ratios (ORs) and 95% CIs using multivariate-adjusted logistic regression. RESULTS Mean BMI was lowest in vegans (23.6 kg/m2) and incrementally higher in lacto-ovo vegetarians (25.7 kg/m2), pesco-vegetarians (26.3 kg/m2), semi-vegetarians (27.3 kg/m2), and nonvegetarians (28.8 kg/m2). Prevalence of type 2 diabetes increased from 2.9% in vegans to 7.6% in nonvegetarians; the prevalence was intermediate in participants consuming lacto-ovo (3.2%), pesco (4.8%), or semi-vegetarian (6.1%) diets. After adjustment for age, sex, ethnicity, education, income, physical activity, television watching, sleep habits, alcohol use, and BMI, vegans (OR 0.51 [95% CI 0.40–0.66]), lacto-ovo vegetarians (0.54 [0.49–0.60]), pesco-vegetarians (0.70 [0.61–0.80]), and semi-vegetarians (0.76 [0.65–0.90]) had a lower risk of type 2 diabetes than nonvegetarians. CONCLUSIONS The 5-unit BMI difference between vegans and nonvegetarians indicates a substantial potential of vegetarianism to protect against obesity. Increased conformity to vegetarian diets protected against risk of type 2 diabetes after lifestyle characteristics and BMI were taken into account. Pesco- and semi-vegetarian diets afforded intermediate protection.